

# Viral Chitlangia

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## Education

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| PhD | Pennsylvania State University, Statistics                                                                                                                                                                    | 2026 -      |
| BS  | Indian Institute of Technology Kanpur, Statistics & Data Science                                                                                                                                             | 2022 - 2026 |
|     | <ul style="list-style-type: none"><li>• CPI : 9.12/10.0</li><li>• Honours Track Student</li><li>• <b>Minor</b> : Dept. of Computer Science &amp; Engineering - Machine Learning &amp; Applications</li></ul> |             |

## Scholastic Achievements

- **Academic Excellence Award, IIT Kanpur** - For exceptional performance in Academic Session 2022-23.
- Was part of the team that represented **IIT Kanpur** in **Mimamsa 2024** hosted by **IISER Pune**, and secured **4th place** among more than **1500** participating teams.
- Achieved All India Rank **646** in **JEE-ADVANCED 2022**
- Achieved All India Rank **1030** in **JEE-MAINS 2022**

## Publications & Pre Prints

- Viral Chitlangia, Mosuk Chow, Sharmishtha Mitra. "Swap Regression Methodology for Predicting Relationship with Historical Bivariate Data." *arXiv preprint arXiv:2508.15479* [🔗](#), August 2025. Aug 2025

## Manuscripts in Preparation

- Viral Chitlangia, Suncica Hadzidedic, Miaoja Fu. "Digital interventions for loneliness: Understanding design elements and features from user reviews and discussions on social media" *Manuscript in preparation* for submission to ICT4AWE 2026.

## Projects

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| <b>SWAP Regression   Prof. Sharmishta Mitra   IIT Kanpur</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Nov '23 - Aug '25 |
| <ul style="list-style-type: none"><li>• Analyzed the paper written by <b>Mosuk Chow</b>, <b>Bing Li</b> and <b>Jackie Q. Xue</b>, on <b>ON REGRESSION FOR SAMPLES WITH ALTERNATING PREDICTORS AND ITS APPLICATION TO PSYCHROMETRIC CHARTS</b> <a href="#">🔗</a> and developed models with alternating predictors on Bivariate Data using the method of SWAP Regression.</li><li>• Defined a new loss function for the <b>Swap Regression</b> model.</li><li>• Implemented ALT-OPT to solve the loss function.</li><li>• Implemented the model to test on real data as <b>US Public Debt and GDP</b>.</li><li>• Applied the model to predict the causality direction of <b>US Public Debt and GDP</b> from 1966 to 2023 without prior knowledge.</li></ul> |                   |
| <b>Digital Intervention for Loneliness   Prof. Suncica Hadzidedic   Durham University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | May '24 - Ongoing |
| <ul style="list-style-type: none"><li>• Explored <b>Reddit</b> and <b>Google</b> to find relevant subreddits and apps which target <b>loneliness</b>.</li><li>• Scraped 700k+ reviews from <b>Reddit</b>, <b>Google Play</b>, <b>Apple Play</b> and <b>A Lonely Life</b> (a loneliness forum), using relevant Python libraries, and Rvest(R), to collect text data on <b>loneliness</b>.</li><li>• Used <b>Topic Modelling</b> on the data collected to <b>cluster</b> the data into relevant top-</li></ul>                                                                                                                                                                                                                                              |                   |

ics, to realise what people are talking about online, to find out which areas of apps people would like to see an improvement in.

#### **Deep Generative Models for Spatial-Temporal Data | Prof. Swapnil Mishra | NUS**

May '25 - Jul '25

- Implementing and analyzing the paper on [AggVAE](#) written by **Swapnil Mishra et al.**
- Working on using the technique to incorporate **AggVAE** with **Population Disaggregation** techniques to get good predictions using Low Resolutions data points.
- Implemented the model on **US Covid Data**, segregated by 9 Regions of the country.

#### **A Review of Enveloping Techniques in Bayesian Statistics | Prof. Dootika Vats | IIT Kanpur**

Aug '25 - Nov '25

- Implemented and evaluated the paper by **Apartim Shukla, Dootika Vats** and **Eric C. Chi** on [MCMC Importance Sampling via Moreau-Yosida Envelopes](#).
- Read about various enveloping techniques in **optimization** and understood the properties of various envelopes, and their usability in MCMC sampling.
- Implemented envelopes, such as Moreau Yoshida and Bregman Moreau Envelopes, for Constraint Distributions.
- Performed Proximal Sampling, using enveloping technique, for distributions with high condition number.

#### **Bayesian Bivariate Load Sharing Model | Prof. Debasis Kundu | IIT Kanpur**

Jan '26 - Ongoing

- Implemented and evaluated the paper, [A Bivariate Load Sharing Model](#) by Debasis Kundu.
- Incorporating a Bayesian structure to the existing model.
- Exploring posterior inference using MCMC-based methods and Gamma-approximate sampling.

#### **MCMC Machine Unlearning | Course Project | Prof. Dootika Vats | IIT Kanpur**

Jan '25 - Apr '25

- Understood and implemented the paper on [Markov chain monte carlo-based machine unlearning: Unlearning what needs to be forgotten](#) by QP Nguyen et al.
- Implemented a novel algorithm for the purpose of Machine Unlearning Sampling by MCMC using the idea of **Newton's Method Update** proposed in [Certified data removal from machine learning models](#) by C Guo Et al.
- Compared the algorithm with other prominent MCMC sampling algorithms with **Logistic** and **Negative Binomial** Regression data.

## **Other Experiences**

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#### **Teaching Assistant | MTH 208 | Prof. Dootika Vats | IIT Kanpur**

Aug '24 - Nov'24

- Assisted Professor Dootika Vats in teaching the Undergraduate and Postgraduate students enrolled in MTH 208 the basics of the R language.
- Got a hands on experience of teaching a class of enthusiastic students.

## **Technical Skills**

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**Languages:** Python, R, C, C++,  $\text{\LaTeX}$

**Softwares and Libraries:** Matplotlib, Seaborn, Numpy, Pandas, Tensorflow, Librosa, Numpyro, Rvest, Tidyverse, GG-Plot2, Quarto, Shiny App

## Relevant Courses

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| An Introduction to Bayesian Analysis*<br>Probability Theory<br>Introduction to Machine Learning*<br>Data Structures & Algorithms<br>Elementary Stochastic Processes - I*<br>Linear Regression & ANOVA*<br>Multivariate Analysis | Markov Chain Monte Carlo*<br>Time Series Analysis<br>Probabilistic Machine Learning<br>Theory of Statistics*<br>Elementary Stochastic Processes - II<br>Techniques in AI & Data Mining*<br>Analysis-I* |
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\* : Excellent Performance

## Positions of Responsibilities

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### Coordinator, Stamatics Club

Oct '24 - Aug '25

- Organized Mathemania 2024 along with a group of 20 executives and 3 sub-coordinators.
- Coordinated with the Mathematics & Statistics Department, IIT Kanpur to secure funding for Stamatics Club in the forthcoming budgets.

### Executive, Stamatics Club

Nov '23 - Oct '24

- Helped organize Mathemania 2023.
- Organized statistics related talks for the campus.